Valerio Dionisi

my website   valerio.dionisi@unimib.it
GitHub   Rome (It), 04/07/1997

RESEARCH INTERESTS

Macroeconomics, Production Networks, Structural Macro-Labour, Fiscal and Monetary Policy

EDUCATION

Visiting Researcher — *University of Oxford*
2024 – 2025

Activities: Production networks, macro-labour, fiscal policy
Advisor: Prof. Francesco Zanetti

Visiting Researcher — *Universitat Pompeu Fabra*
2023 – 2024

Activities: Numerical techniques for quantitative macroeconomics, production networks
Advisor: Prof. Davide Debortoli

Ph.D. in Economics, Statistics and Data Science (*cum laude*) — *University of Milano-Bicocca*
2021 – 2026

Thesis: "Macroeconomic Perspectives on Sectoral Composition: Networks and Inequality"
Committee: Basile Grassi (Bocconi), Alessio Moro (UniCa), Francesco Zanetti (Oxford), Enzo Dia (UniMiB)
Supervisor: Prof. Andrea Colciago

Ms.C. in European Economy and Business Law (*cum laude*) — *Tor Vergata University*
2019 – 2021

Thesis: "Evidence of Debt Consolidation Dynamics for Europe and Italy"
Supervisor: Prof. Barbara Annicchiarico, *Co-supervisor*: Prof. Leonardo Becchetti

B.A. in Political Sciences — *Roma Tre University*
2016 – 2019

Thesis: "Instrument-Target Alignment and the Coordination of Economic Policies"
Supervisor: Prof. Gian Cesare Romagnoli

Further Education

Advanced Computational Macroeconomics, 2022 — *London School of Economics*
Mathematics for Social Sciences, 2018 — *Roma Tre University*

RESEARCH PAPERS

The Horizontal Geometry of Production Networks — GO TO THE PAPER
2026, Job Market Paper

(***abstract***). Modern economies are organized as dense production networks, yet comovement is typically understood through observed Input-Output linkages and vertical cascade effects. This paper argues that this view is incomplete. Comovement fundamentally depends on horizontal complementarities: the extent to which sectors share upstream suppliers or downstream buyers. Such network similarities in demand and supply induce simultaneous responses to idiosyncratic shocks, shaping the sign of comovement. Regarding aggregate fluctuations, horizontal transmission generates systematic comovement that prevents shocks from averaging out, without relying on cascading amplification. The framework developed highlights this novel diffusion channel as a complement to vertical propagation, offering a unified view of comovement rooted in the horizontal geometry of common economic interactions rather than network trade intensity. Using Input-Output data, I show that sectoral employment responses vary systematically with horizontal distances, with nearby sectors exhibiting dampened or opposing responses, and distant ones comoving positively.

Networks, Real Rigidities, and Monetary Policy DRAFT UPON REQUEST
2026, with Colciago A. (DNB, UniMiB), Siena D. (PoliMi, Bocconi), and Zago R. (BdF, CdF, ESCP)

(***abstract***). Over the past decades, the slope of the Phillips Curve has exhibited substantial variation across countries and sectors, with a marked flattening prior to the pandemic. This paper studies how the topology of domestic and global production networks shapes inflation dynamics. Using country-industry data on producer prices and output, we document systematic heterogeneity in Phillips-curve slopes: downstream sectors and industries with strong backward integration display flatter inflation responses, while upstream sectors and industries with strong forward linkages exhibit steeper dynamics. We rationalize these asymmetries with a network-based DSGE model where the NKPC slope reflects the interaction between real rigidities (strategic complementarities and endogenous markups), network propagation of nominal rigidities, and global slack transmission through imported intermediates. The results imply that the aggregate Phillips Curve slope is an endogenous object that shifts with production network structure and global integration.

Industrial Structure, Technological Change, and U.S. Wage Inequality GO TO THE PAPER
2026

(***abstract***). This paper examines the determinants of U.S. wage inequality between industries. It distinguishes two channels of Skill-Biased Technological Change: a quantity effect, mirroring variations in the distribution of capital and labour inputs, and a structural effect, reflecting differences in substitution elasticities across factors of production. Counterfactual exercises show that evolving technological heterogeneity in substitution patterns dominates wage dispersion. A quantitative decomposition reveals that structural forces, mostly reflecting substitutability of routine with non-routine labour, account for 94% of observed wage variance, exacerbated by stronger sorting and segregation effects. Upon neutralising structural differences, the quantity channel reckons merely 6-15%. Indeed, structural heterogeneity, rather than mere changes in factor quantities, is the dominant force behind U.S. wage inequality.

WORK IN PROGRESS

Vertically-Integrated Networks and the Consumption Channel

Where to Sit in the Chain: Microfoundations and Micro-to-Macro Implications of Global Value Chain Positioning, *with Siena D. (PoliMi, Bocconi)*

Job Vacancies through COVID-19 Lockdowns in the Euro Area, *with Bertucci A. (UniPv)*

TEACHING

2024/2025-*still*	International Macroeconomics (Grad), *Lecturer*	*Polytechnic University of Milan*
2024/2025-*still*	Macroeconomics (UndGrad), *Lecturer*	*Polytechnic University of Milan*
2023/2024-*still*	Computational Macroeconomics (Grad), *Lecturer*	*University of Milano-Bicocca*
2023/2024-*still*	Dynamic Asset Pricing (Grad), *TA*	*University of Milano-Bicocca*
2023/2024-*still*	Financial Contract Theory (Grad), *TA*	*University of Milano-Bicocca*
2023/2024-*still*	Macroeconomics (UndGrad), *TA*	*University of Milano-Bicocca*

SKILLS

Languages:	ITALIAN (native), ENGLISH (fluent), FRENCH and SPANISH (intermediate)
Programming:	MATLAB, DYNARE, STATA (advanced); JULIA, R (proficient); LaTeX (advanced)
Microsoft Office:	WORD, EXCEL, POWERPOINT, OUTLOOK

Certificates

2021	Introduction to Linear Algebra, Introduction to Statistical Methods	*MatLab*
2021	Fundamentals, OnRamp, Optimization OnRamp	*MatLab*

CONFERENCES AND SEMINARS, *INCLUDING SCHEDULED*

2026

4th Milan Ph.D. Economics Workshop, *Organizer*; 25th Research in International Economics and Finance Network Workshop (***scheduled***), International Association for Applied Econometrics (IAAE) Annual Conference (***scheduled***), Irish Economic Association Annual Conference (***scheduled***), 1st Piraeus International Economic Conference (***scheduled***), Tor Vergata University (***scheduled***), Roma 3 University, Catholic University of Milan, University of Milano-Bicocca, *Presenter*

2025

3rd Milan Ph.D. Economics Workshop, *Organizer*; University of Oxford, Naples School of Economics: 4th PhD and Post-Doctoral Workshop, University of Naples "Federico II", "Sapienza" University of Rome, Catholic University of Milan, University of Milano-Bicocca, *Presenter*

2024

2nd Milan Ph.D. Economics Workshop, *Organizer*, University of Pavia, Catholic University of Milan, University of Milano-Bicocca *Presenter*

2023

1st Milan Ph.D. Economics Workshop, University of Milan, University of Milano-Bicocca, Catholic University of Milan, *Presenter*

REFERENCES

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